HR Analytics Dashboard

Project Report

1. Introduction

The HR Analytics Dashboard is a business intelligence project designed to analyze employee demographics, attrition, salary, job satisfaction and tenure. The dashboard converts cleaned employee-level data into interactive KPIs and visualizations that help identify workforce trends and potential attrition patterns.

2. Objectives

• Measure employee count and attrition.

• Calculate and display the attrition rate.

• Analyze attrition across gender, age group, education field, job role and salary slab.

• Study the relationship between job satisfaction and attrition.

• Understand attrition across years spent at the company.

• Provide a department-level interactive filter.

3. Tools and Technologies

Power BI Desktop, Power Query, DAX, Python, Pandas, Plotly and Streamlit were used. Power BI is the original report platform; the Python implementation provides a code-based interactive version for source-code submission.

4. Dataset

The cleaned dataset contains employee identifiers, age and age groups, department, education field, gender, job role, job satisfaction, monthly income, salary slab, attrition status, attrition count and years at company, along with additional HR attributes.

5. Dashboard Components

Component	Purpose
Employee Count	Shows the number of employees in the current filter context.
Attrition	Shows the number of employees marked as attrited.
Attrition Rate	Shows attrition as a percentage of filtered employees.
Average Age	Shows average employee age.
Average Salary	Shows average monthly income.
Years At Company	Shows average tenure at the company.
Education Field	Compares attrition across education fields.
Age Group	Compares attrition across age groups.
Gender	Shows attrition by gender.
Job Role	Ranks job roles by attrition.
Salary Slab	Shows attrition by salary range.
Years At Company	Shows attrition trend by tenure.
Job Satisfaction	Provides a job-role and satisfaction-level matrix.

6. Methodology

The cleaned CSV is loaded into the Python application with Pandas. Numeric columns are converted to numeric types, derived attrition counts are used where required, and the selected department is applied as a filter before KPI and chart calculations. Plotly generates the interactive charts and Streamlit provides the dashboard interface.

7. Expected Outcome

The final dashboard provides a single view of workforce health and attrition patterns. Users can select a department and immediately see how the key metrics and visualizations change.

8. How to Run the Python Version

Open the project folder in VS Code, create a virtual environment, install the packages in requirements.txt, and run **streamlit run app.py**. The dashboard opens in a browser.

9. Files Submitted

• app.py – Python dashboard source code

• requirements.txt – Python dependencies

• HR_Analytics_Final_Cleaned.csv – cleaned dataset

• HR_Dashboard.pbix – original Power BI dashboard

• README.md – project documentation

• Project_Report.pdf – project report

10. Conclusion

The HR Analytics Dashboard demonstrates how cleaned employee data can be transformed into meaningful HR insights using business intelligence and Python visualization tools. It can support management in identifying high-attrition groups and understanding workforce characteristics.